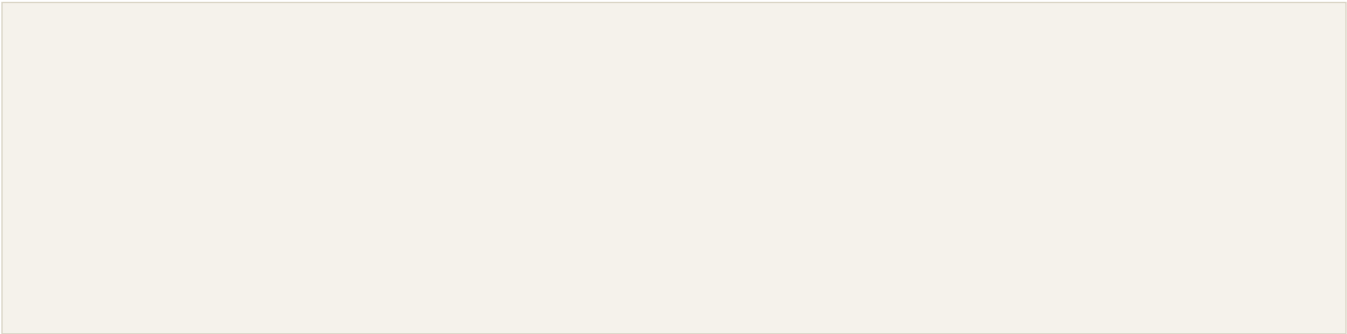


This document defines the first practical architecture for the Pattern App brain.

The brain is not one prompt. It is a staged reasoning system that turns practitioner intake into cited, tradition-separated interpretations, then synthesizes across traditions while preserving uncertainty and safety.

The app ranks source-supported traditional-system relevance, not medical truth.

It must never diagnose, prescribe, or collapse Ayurveda, TCM, and Homeopathy into one blended system.



Safety runs before every other reasoning step.

Inputs:

- ? symptoms
- ? severity
- ? duration
- ? pregnancy status
- ? medications
- ? known conditions
- ? red-flag language

Outputs:

- ? red_flags_detected
- ? safety_level
- ? suppressed_traditional_suggestions
- ? referral_language

Behavior:

- ? If emergency red flags appear, suppress herbs, formulas, remedies, yoga, breath, and lifestyle suggestions.
- ? If high-priority medical review signs appear, lower confidence and add practitioner review warnings.
- ? If medication/pregnancy/condition data is missing, apply confidence penalties.

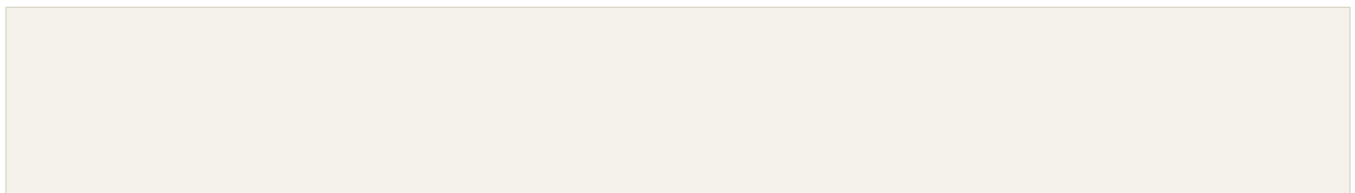
The normalizer converts practitioner notes into structured features without making conclusions.

Feature groups:

- ? chief complaint
- ? primary symptoms
- ? secondary symptoms
- ? onset, duration, severity
- ? location, quality, timing
- ? better/worse modalities
- ? temperature, thirst, sweat
- ? digestion, stool, urine
- ? sleep, energy, mood
- ? constitution/context
- ? medications, pregnancy, allergies, known conditions
- ? tradition-specific observations

Normalized features should preserve the user's language and also map to controlled concept tags.

Example:



Retrieval must run separately for each tradition.

Ayurveda retrieval searches:

- ? dosha patterns
- ? agni and ama language
- ? dhatu/mala/srotas concepts
- ? classical citations
- ? modern assessment frameworks when private/local access exists

TCM retrieval searches:

- ? organ-system patterns
- ? hot/cold
- ? excess/deficiency
- ? qi/blood/fluid
- ? damp/phlegm/stagnation
- ? formula and herb logic

Homeopathy retrieval searches:

- ? symptom totality
- ? characteristic symptoms
- ? modalities
- ? mental/emotional generals
- ? remedy pictures
- ? repertory rubrics
- ? confirmatory symptoms

Retrieval output must include:

- ? source chunk ID
- ? source title
- ? tradition
- ? page/chapter/aphorism/rubric locator
- ? text preview
- ? matched features
- ? citation quality score
- ? rights/access note

The candidate generator groups retrieved chunks into possible interpretations.

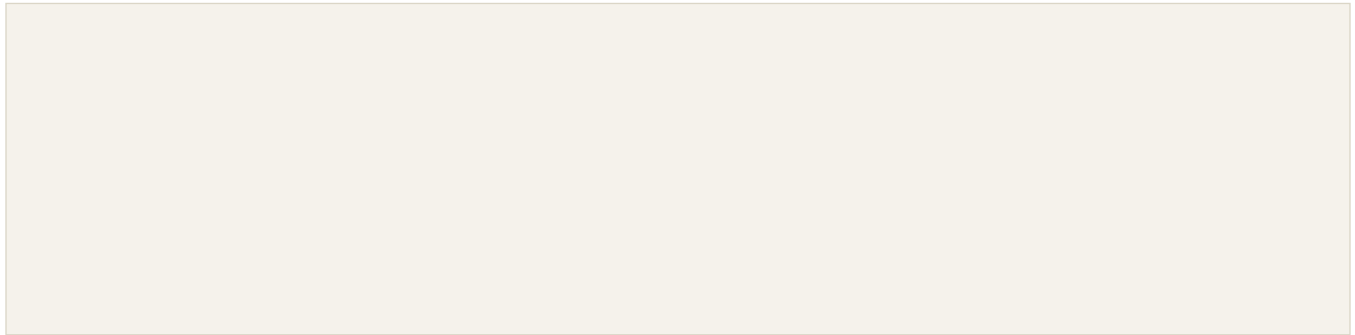
Examples:

? Ayurveda candidate: vata aggravation with dry/irregular digestion features

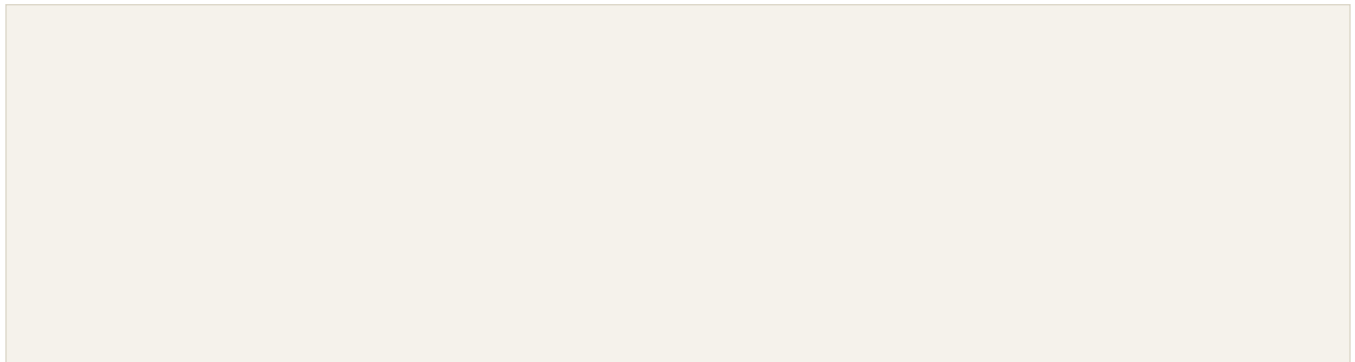
? TCM candidate: spleen qi deficiency with dampness

? Homeopathy candidate: remedy differential: Nux vomica / Lycopodium / Pulsatilla

Each candidate should include:



The first scoring formula:



Scores should be explainable. Every output should be able to answer:

? What matched?

? What did not match?

? What contradicted the candidate?

? Which citations support it?

? What missing answer would change confidence most?

Confidence labels:

- ? 85-100: strong source-supported match
- ? 70-84: likely match, practitioner review required
- ? 50-69: possible match, needs more intake detail
- ? 30-49: weak match, exploratory only
- ? 0-29: insufficient evidence

No output should make a source-backed claim without a citation object.

Citation fields:

- ? citation_id
- ? source_id
- ? title
- ? author
- ? translator
- ? edition
- ? tradition
- ? page_start
- ? page_end
- ? chapter
- ? sutra_or_aphorism
- ? rubric
- ? source_url
- ? rights_note

The citation engine should distinguish:

- ? public extracted source
- ? private/local-only source
- ? metadata-only source
- ? excerpt-only public preview

Synthesis happens only after tradition-specific analysis.

It compares reasoning dimensions, not labels:

- ? heat/cold
- ? excess/deficiency
- ? stagnation/flow
- ? dryness/dampness
- ? depletion/agitation
- ? constitution/predisposition
- ? modalities
- ? intervention rationale

Synthesis output:

- ? shared themes
- ? areas of agreement
- ? areas of conflict
- ? non-equivalence notes
- ? what needs practitioner review
- ? confidence score

Forbidden synthesis behavior:

- ? saying traditions are identical
- ? claiming scientific proof
- ? using one tradition to validate another
- ? turning suggestions into prescriptions

The app can suggest categories for practitioner review, not prescriptions.

Allowed category language:

- ? Ayurveda: herbs, formulas, diet, lifestyle, yoga/breath practices
- ? TCM: herbs, formulas, diet, lifestyle
- ? Homeopathy: remedy differentials, repertory rubrics, modalities, constitution notes

Output should say:

- ? suggested categories for practitioner review
- ? source-supported direction
- ? requires qualified review

Output should not say:

- ? take this
- ? prescribe this
- ? this will cure
- ? diagnosis is

The app should have two kinds of memory.

Project memory:

- ? source registry
- ? chunk schema
- ? known citations
- ? synonym maps
- ? pattern maps
- ? remedy/rubric maps
- ? safety rules

Case memory:

- ? current intake
- ? missing information
- ? questions already asked
- ? practitioner corrections
- ? outcome/follow-up notes

Do not let case memory overwrite source truth. Corrections should update the case state, not mutate source data.

1. Finish public Homeopathy retrieval layer:

- ? Boericke materia medica
- ? Boericke repertory
- ? Kent repertory
- ? Kent lectures

2. Add controlled concept tags:

- ? symptoms
- ? modalities
- ? temperature
- ? digestion
- ? sleep
- ? mental/emotional generals
- ? red flags

3. Build candidate generator:

- ? group citations by pattern/remedy/rubric cluster
- ? produce matched/unmatched/contradicting features

4. Build scoring engine:

- ? transparent score breakdown
- ? missing-info penalties
- ? contradiction penalties

5. Build next-best-question engine:

- ? ask the one question most likely to change the ranking

6. Build synthesis engine:

- ? compare dimensions after tradition-specific scoring
- ? show agreement and conflict without flattening traditions

7. Add private-source support:

- ? Vasant Lad
- ? Vithoulkas
- ? Sankaran
- ? Morrison
- ? private extracted text stays gitignored

The strongest architecture is a hybrid system:

- ? structured rules for safety, metadata, citations, and scoring
- ? lexical/vector retrieval for finding relevant source passages
- ? LLM summarization only after evidence is retrieved

? deterministic output schema validation before anything reaches the UI

Do not make the LLM the source of truth. Make the LLM the explainer and comparator after the retrieval/scoring system has assembled evidence.

That gives the app a calm, intelligent voice without letting it hallucinate the medical/traditional reasoning.

The first testable brain trace is available in:

? src/pattern_app_brain.py

? app/pattern-app/page.tsx

? app/api/pattern-brain/route.ts

Run the CLI trace:

Run the local browser prototype:

Then open: